AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

Test Report No. 2023 – 0015

Agricultural Machinery – Power Disc Harrow SHANDONG YUNTAI 1LYQ-622

College of Engineering and Agro-Industrial Technology, University of the Philippines Los Baños, College, Laguna 4031
Email: amtec.uplb@up.edu.ph Facebook: www.facebook.com/amtec.uplb Website: www.amtec.ceat.uplb.edu.ph
Contact Numbers: +63 921 400 7137 (Smart) / +63 917 704 0922 (Globe)

Test Report Number	2023 – 0015
Date of Issuance	February 16, 2023
Valid Until	February 16, 2028
Agricultural Machinery	Power Disc Harrow
Type	Tractor-mounted, Single-action, PTO-driven
Brand	SHANDONG YUNTAI
Model	1LYQ-622
Test Requested by	FORD TRACTOR PHILS. INC 1384, Jose Abad Santos, Tondo, Manila

SUMMARY

Construction Criteria	Requirement as per PAES 120:2001	AMTEC Test
Length, mm	2000 to 8000	1760
Width, mm	1000 to 10000	1845
Height, mm	1000 to 1500	1150
Number of Gangs	Minimum of 2 (offset and single-action disc harrow)	1
Type of Disc	Plain or notched	Plain
Number of Discs Per Gang	Minimum of 4	6
Diameter of Discs, mm	400 to 650 (600 maximum for four-disc gang)	555
Thickness of Disc, mm	4 to 6	5.01
Concavity (Depth), mm	100 to 300	67
Disc Angle, °	24 maximum	23.56
Length of Spool, mm	175 or 225 ± 2	225
Ground Clearance, mm	125	180
Weight Per Disc, kg	25 to 70	NM
Drawbar Power Requirement Per Disc, kW	1 to 2.5	5.10[a]

[a] Drawbar power requirement is computed as 86% of the maximum PTO power from the YTO MF554 performance test TR 2020 – 0209 divided by the number of discs.
Note: NM – Not Measured

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not Provided
e. Maintenance instruction	Provided
f. Storage	Not provided
g. Handling, reception, transportation, assembly and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	SHANDONG YUNTAI MACHINERY CO., LTD. South Weier Road, West of Mingjia Dong Road, Qihe Economy Development Zone, Shandong, China
4. Four-wheel tractor used	40.4 kW YTO MF554
5. Serial number	*42211185*
6. Number of operators	One
7. Handling and stability when machine is working	The disc harrow hitched to the tractor was stable during operation.
8. Handling and stability when machine is turning	The disc harrow hitched to the tractor was stable. It was lifted using the hydraulic cylinder during turning.
9. Quality of finished field: a. Tilth b. Level	The soil was tilled, and the field was leveled after harrowing.
10. Non-adhesion of soil to disc	Minimal adhesion of soil to the discs was observed.
11. Replacing and adjusting of parts	Some parts can be easily replaced and adjusted. No parts were replaced and adjusted during the test operation.
12. Durability of parts (based on wear of soil-working parts, visible deformation, etc.)	There was no visible deformation on the components during and after the test operation.
13. Safety features	Power transmission systems have guards.
14. Failure or abnormalities that may be observed on the tractor or its component parts	No breakdown occurred during the test operation.

OBSERVATIONS (Cont'n)

15. Mild steel shall be used in the manufacture of the frame, gang axle, scraper, gang angling mechanism, and hitch.	Channel bar, square tube (frame), and round bar (gang axle) are used to manufacture the machine's components
16. Cast iron shall be used in the manufacture of spool.	It was not verified if cast iron was used in the manufacture of the spool.
17. Carbon steel shall be used in the manufacture of hitch pin.	It was not verified if carbon steel was used in the manufacture of the hitch pin.
18. Carbon steel with at least 80% carbon content (e.g. AISI 1080) or alloy steel with at least 0.0005% boron content shall be used in the manufacture of the disc blades.	It was not verified if carbon steel with at least 80 % carbon or alloy steel with at least 0.0005 % boron content was used in the manufacture of the disc blades.
19. The frame shall be rigid and durable.	The frame is rigid and stable.
20. The gangs shall be so attached that their angles can be easily changed to desired position with gang angling mechanism.	The gang angle cannot be changed.
21. The rear gang(s) in offset and tandem disc harrows shall be so arranged that the disc(s) of rear gang(s) shall not make the same path formed by front gang(s).	Not applicable
22. Bumpers may be provided in the inside end of the front gangs of the single-action and tandem disc harrows to prevent the inside discs from rubbing each other at any gang angle in which the gangs are set.	Not applicable
23. The discs in each gang shall be firmly fixed by spools and there shall not be any relative movement between the disc and the axle.	A spool is placed between the discs to prevent lateral movement of the shaft.
24. The scrapers shall be set in such a way that they shall not touch the face of disc and shall be able to scrape the soil effectively. Arrangement for adjusting the scrapers shall be provided.	Scrapers are not present in the machine.
25. Adequate arrangement for lubrication of bearings shall be provided. The bearings shall be reasonably dust-proof and shall be properly aligned.	The bearings are provided with lubrication points.
26. Ball or tapered roller bearings with special dirt seals shall be used	Ball bearings are used in the machine.
27. Both faces of spool coming in contact with the disc face shall be finished for proper gripping.	The spools in between discs are properly finished. They prevent lateral movement of the discs.

OBSERVATIONS (Cont'n)

28. The hitch of disc harrow shall be compatible with the hydraulic system and the three-point hitch of the tractor specified in PAES 118.	The drawbar hitch of the disc harrow is compatible with that of the tractor
29. In single-action disc harrow, provision may be made for attaching a shovel or a sweep to cut and uncut the soil between the two gangs.	Not applicable
30. The harrow shall be easy to operate such as:	
a. Hitching to and unhitching from tractor;	Hitching and unhitching was easy.
b. Adjusting the depth of cut;	The depth of cut can be adjusted through the three-point linkages.
c. Changing the position of the harrow with respect to the line of pull of the tractor;	Not observed
d. Moving the standards laterally on the frame;	Not applicable
e. Varying the angle between the bodies;	The gang angle cannot be adjusted.
f. Maneuverability during operation;	The power disc harrow hitched to the tractor was stable when turning. The power disc harrow was lifted using the three-point linkage during turning.
g. Clearing blockages;	No blockages were observed during test operation.
h. Changing from transport to work position and vice versa; and	Changing the position from transport to work and vice versa can be done through the three-point linkages.
i. Adjustment of the scrapers.	Scrapers are not present in the implement.
31. Others	a. No markings and labels were provided. b. A furrow wheel was provided to counteract the force of the single gang harrow. c. The discs are powered by the PTO to rotate and remove any soil adhesion. d. The gang was angled at 23.56° with respect to the line perpendicular to the direction of travel.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 18, 2023
Location of Test	Luisita Access Road, Tarlac City, Tarlac 15°26'08.36" N, 120°39'12.56" E
Items Tested	Operator's Manual, Depth of Harrowing, Width of Harrowing, Field Capacity, Field Efficiency, Noise Level, Number of Operators, and Fuel Consumption Rate

METHODS OF TEST

The disc harrow was tested following the PAES 133:2004 – Agricultural Machinery – Disc Harrow – Methods of Test.

DESCRIPTION OF THE IMPLEMENT

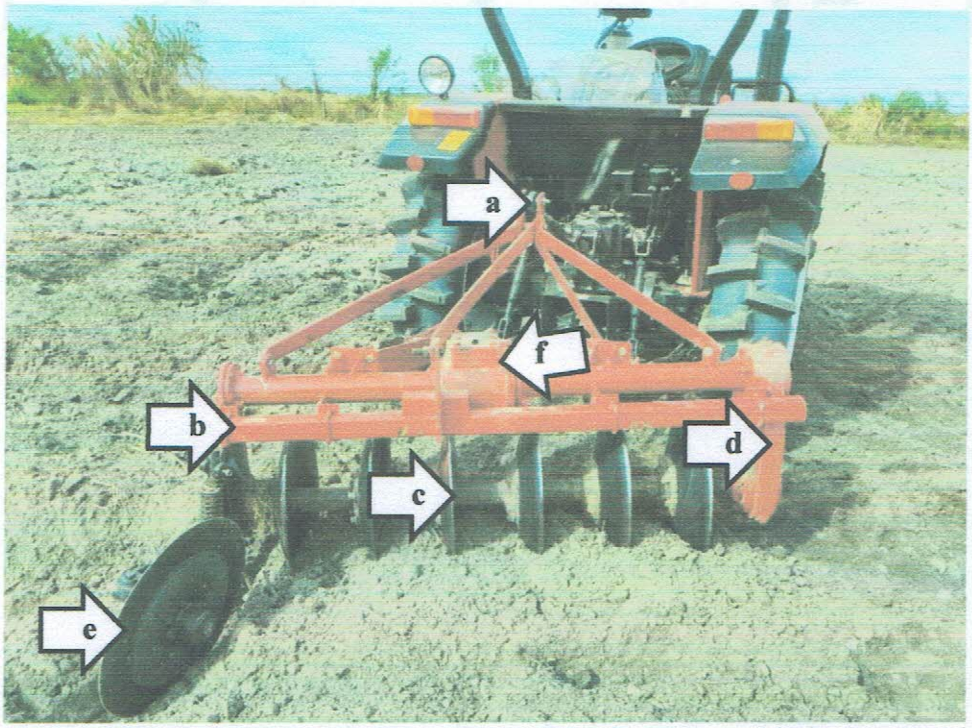

Figure 1. Parts of SHANDONG YUNTAI 1LYQ-622 Disc Harrow hitched to 40.4 kW YTO MF554 include the following: (a) Upper link hitch point, (b) Main frame, (c) Discs, (d) Side drive with guard, (e) Rear thrust wheel, and (f) Gearbox.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification
1	Overall dimensions and weight		
1.1	Length, mm	1210	1760
1.2	Width, mm	1805	1845
1.3	Height, mm	1050	1150
1.4	Weight, kg	321	NM
1.5	Weight per disc, kg	ND	NM
2	Ground clearance, mm	ND	180
3	Gang		
3.1	Quantity	ND	1
3.2	Number of discs per gang	ND	6
3.3	Disc spacing, mm	ND	248
3.4	Angle of gang, °	ND	23.56
3.5	Size and shape of gang shaft	ND	NM
4	Disc		
4.1	Brand	SHANDONG YUNTAI	SHANDONG YUNTAI
4.2	Make	SHANDONG YUNTAI MACHINERY CO., LTD. China	SHANDONG YUNTAI MACHINERY CO., LTD. China
4.3	Type	ND	Plain
4.4	Diameter, mm	ND	555
4.5	Thickness, mm	ND	5.01
4.6	Concavity, mm	ND	67

[a] The data were obtained from the manual provided by the requesting party.
Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification
5	Scraper	NA	NA
6	Spool		
6.1	Length, mm	ND	225
6.2	Shape and size of hole, mm	ND	NM
6.3	Material	ND	ND
7	Main frame		
7.1	Dimension, L × W, mm	ND	1725 × 1190
7.2	Material	ND	ND
8	Transport wheels	NA	NA
9	Bearings		
9.1	Type	ND	Ball bearing
9.2	Brand	ND	ND
9.3	Quantity per gang	ND	2
10	Additional weight, kg	ND	None
11	Operating width, mm	ND	1310

[a] [a] The requesting party did not submit complete machine specifications

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Test conditions		
1.1	Type of field operation	Secondary – Harrowing	
1.2	Condition of field		
1.2.1	Location	Luisita Access Road, Tarlac City, Tarlac	
1.2.2	Dimensions of field, L × W, m	46.0 × 23.0	
1.2.3	Area, m^2	1058.0	
1.2.4	Soil type	Sandy loam	
1.2.5	Soil moisture content, $\%_{db}$	7.14	
1.2.6	Height of stubbles, mm	NA	
1.2.7	Spacing of stubbles, row × hill, mm	NA	
1.2.8	Weed density	NA	
1.2.9	Soil hardness, kg/cm^2	NA	
1.2.10	Last crop planted	Mungbean	
1.2.11	Field condition before operation	Plowed	
2	Field performance	1^{st} pass	2^{nd} pass
2.1	Brand and model of tractor used	YTO MF554	
2.2	Tractor's gear shift setting	4^{th} gear – Low	
2.3	Traveling or operating speed, kph	8.91	7.47
2.4	Average depth of harrowing, mm	202	180
2.5	Average width of harrowing, mm	1355	1210
2.6	Operating width, mm	1310	1310
2.7	Actual field capacity		
2.7.1	ha/h	0.738	0.596
2.7.2	h/ha	1.36	1.68
2.8	Theoretical field capacity		
2.8.1	ha/h	1.17	0.979
2.8.2	h/ha	0.856	1.02
2.9	Field efficiency, %	64.27	60.55
2.10	Fuel consumption rate, L/h	7.82	10.73
2.11	Specific fuel consumption, L/ha	10.54	18.08
2.12	Field operational pattern	Modified circuitous pattern rounded corners	
2.13	Unharrowed, %	8.87	14.57 (overlapped)
2.14	Noise level, dB(A)	90.2	90.0

Note: NA – Not Applicable

OBSERVATION

Figure 2. Quality of work

Tested by:

RHEY MARC C. ALBALOS	**CHRISTOPHER S. JAPONES**
Test Engineer	Test Engineer

Verified by:

MARIE JEHOSA B. REYES
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO	FEB 20 2023
AMTEC Director	Date

REMINDER

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